TTG+

From native traversal continuations to fused SQL and ordered cache entries

LinkedList32 · LinkedList1000 · original LittleX

Off · repaired pre-fusion process · TTG+ process

36 fresh workflow trials: 9 excluded warmups + 27 measured

Actual DB protocol time, total SQL, native correctness and process/backend intervals

The problem: native traversal crosses DB boundaries

The walker preserves application order; each demanded next step can require path and payload access.

```
walker Traverse {
  has visited: int = 0;
  can visit_item with Item entry {
    self.visited += 1;
    report here.value;
    if self.visited >= 10000 { disengage; }
    visit [->:Next:->];
  }
}
```

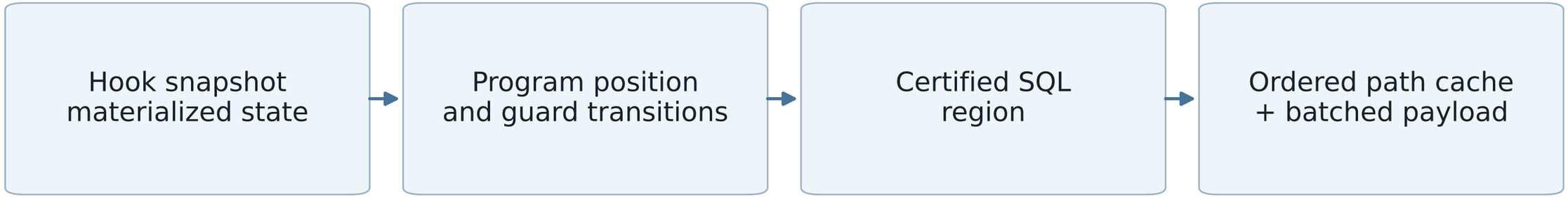


Item —Next→ Item —Next→ ...

Ordered reports and visited=N remain native correctness checks.

Mechanism: snapshot → certified region → native cache

Same repaired runtime; the explicit ttg_fused_sql switch selects the fused branch.



Native execution consumes the cached path in its original order.
Unsupported or unsafe tails retain their recorded fallback reason.

Why is a second visit guaranteed?

The analyzer supplies a conditional control-flow certificate; SQL checks the actual graph facts.

Same LinkedList, at the first visit hook

```
origin = A : Item
ability/event = Traverse.visit_item / visit
scalar snapshot: visited = 1
cutoff: visited < 10000
pending FIFO = []
one known Item-entry handler
rule: here --Next--> Item
max_origins encodes cutoff + budgets
```

Analyzer proof



Illustrative four-node instance; measured sizes are 32 and 1,000.

After visit(A) returns one Item,
FIFO dispatch enters that same ability at B.
At depth d, visited = 1 + d; the certified bound stays before disengage. SQL consumes that bound, without interpreting Jac code.

The code certificate does not assert that an edge exists. SQL must observe exactly one successor

of the exact certified runtime type/module before recursing. Zero or multiple successors stop linear expansion;

a nonempty pending FIFO also prevents this linear certificate. Native execution still handles the actual path.

Graph evidence in SQL → ordered native cache

Simplified from the final SQL AST contract; native edge/type/ACL/order predicates and bound parameters are abbreviated.

```
WITH RECURSIVE walk(origin_id, depth) AS (
  VALUES (:origin, 0)
  UNION ALL
  SELECT m.target_id, w.depth + 1
  FROM walk w, LATERAL native_path(w.origin_id) m
  WHERE m.arity = 1
    AND (m.arch_type, m.arch_module) = (:type, :module)
    AND w.depth < :max_origins - 1
)
SELECT w.origin_id, w.depth, p.target_id, p.ord, p.arity
FROM walk w LEFT JOIN LATERAL native_path(w.origin_id) p
  ON TRUE
ORDER BY w.depth, p.ord;
```

Per-origin ordered path cache

```
(A, Next) → [(B, edgeAB)]
(B, Next) → [(C, edgeBC)]
(C, Next) → [(D, edgeCD)]
(D, Next) → []
```

The empty terminal result is cached too.
Native visit consumes its own path order.
Payload IDs are merged for process loading.
SQL discovers paths; the walker still runs.

LittleX: two proven visits in the current collect ability

```
collect(Profile):  visit Post → Tweet;
                   visit Follow → Profile → Post → Tweet;
gather(Tweet):     self.feed.append(here.to_view());
```

Both current-ability path rules can be batched.
Queued gather runs later; to_view is a barrier for proving further ability effects.

LinkedList · 32 nodes: DB time and total SQL

Median [minimum–maximum] across three independent measured fresh workflows

Configuration	DB sum ms	DB union ms	Total/data/control/other SQL
Off	114.325 [114.149–122.103]	114.325 [114.149–122.103]	360/318/42/0
Repaired pre-fusion / process	121.755 [121.182–136.588]	121.755 [121.182–136.588]	365/318/47/0
TTG+ / process	94.459 [94.404–101.485]	94.459 [94.404–101.485]	243/196/47/0

Primary scope: business workflow. Setup, authentication, import, steady reads and verification are retained separately.

Sum adds all foreground/background DB calls. Union merges their actual protocol intervals.

LinkedList · 1,000 nodes: DB time and total SQL

Median [minimum–maximum] across three independent measured fresh workflows

Configuration	DB sum ms	DB union ms	Total/data/control/other SQL
Off	3310.437 [3130.876–3312.781]	3310.437 [3130.876–3312.781]	9072/9030/42/0
Repaired pre-fusion / process	3481.763 [3448.737–3500.474]	3481.763 [3448.737–3500.474]	9077/9030/47/0
TTG+ / process	2336.291 [2300.719–2359.249]	2336.291 [2300.719–2359.249]	5083/5036/47/0

Primary scope: business workflow. Setup, authentication, import, steady reads and verification are retained separately.

Sum adds all foreground/background DB calls. Union merges their actual protocol intervals.

LittleX · original workflow: DB time and total SQL

Median [minimum–maximum] across three independent measured fresh workflows

Configuration	DB sum ms	DB union ms	Total/data/control/other SQL
Off	41.426 [37.734–41.728]	41.426 [37.734–41.728]	153/90/63/0
Repaired pre-fusion / process	49.324 [47.245–51.183]	49.324 [47.245–51.183]	159/90/69/0
TTG+ / process	55.435 [55.168–59.843]	55.435 [55.168–59.843]	147/78/69/0

Primary scope: business workflow. Setup, authentication, import, steady reads and verification are retained separately.

Sum adds all foreground/background DB calls. Union merges their actual protocol intervals.

LinkedList: native Traverse counts and all-role SQL

One actual Traverse request per fresh workflow; full workflow also includes Sum and four Find operations.

Workload	Configuration	Visited	Total SQL [range]	Data SQL	Control SQL
LinkedList · 32 nodes	Off	32	76 [76–76]	69	7
LinkedList · 32 nodes	Repaired pre-fusion / process	32	77 [77–77]	69	8
LinkedList · 32 nodes	TTG+ / process	32	16 [16–16]	8	8
LinkedList · 1,000 nodes	Off	1000	2012 [2012–2012]	2005	7
LinkedList · 1,000 nodes	Repaired pre-fusion / process	1000	2013 [2013–2013]	2005	8
LinkedList · 1,000 nodes	TTG+ / process	1000	16 [16–16]	8	8

Total counts retain path, payload and control SQL across foreground and background roles.

LinkedList: Traverse protocol time

Median [minimum–maximum] ms; the sum and union are equal in every measured Traverse sample.

Workload	Configuration	DB sum = union ms	Total SQL
LinkedList · 32 nodes	Off	25.506 [24.638–26.753]	76
LinkedList · 32 nodes	Repaired pre-fusion / process	26.315 [25.098–26.899]	77
LinkedList · 32 nodes	TTG+ / process	6.971 [6.509–9.869]	16
LinkedList · 1,000 nodes	Off	713.524 [708.183–760.602]	2012
LinkedList · 1,000 nodes	Repaired pre-fusion / process	771.718 [760.090–781.492]	2013
LinkedList · 1,000 nodes	TTG+ / process	31.792 [31.297–67.288]	16

Native ordered reports are 0…31 and 0…999, with visited=32 and visited=1,000 respectively.

LinkedList: both path and payload calls collapse

Actual Traverse SQL classifications; every value is identical in all three measured repeats.

Size	Configuration	Ordinary path	Fused path	Payload	Control	Identity + catalog	TOTAL
LL 32	Off	32	0	35	7	2	76
LL 32	Repaired pre-fusion / process	32	0	35	8	2	77
LL 32	TTG+ / process	0	1	5	8	2	16
LL 1000	Off	1000	0	1003	7	2	2012
LL 1000	Repaired pre-fusion / process	1000	0	1003	8	2	2013
LL 1000	TTG+ / process	0	1	5	8	2	16

Per-origin ordered path entries replace repeated path discovery; one process payload batch replaces repeated loads.

The totals include both roles, the type/module catalog proof, identity lookup and transaction controls.

Limited coalescing: Find keeps its data-dependent branch

LinkedList1,000 · actual SQL per operation; all three measured repeats have the same counts.

Native operation	Off	Pre-fusion	TTG+
Traverse	2012	2013	16
Sum	2012	2013	16
Find head	12	12	12
Find middle	1013	1014	1014
Find tail	2011	2012	2012
Find missing	2012	2013	2013

Native Find guard

```
if here.value == self.target {
  self.found = True;
  report here;
  disengage;
}
visit [->:Next:->];
```

Observed planner stop: Unknown scalar fact: 'visit_item::here.value'

Middle/tail/missing Find issue 500/999/1,000 fused path queries, each caching one origin.

Traverse and Sum coalesce; these Find requests preserve per-step path and payload counts.

LittleX: fewer calls, higher protocol contribution

Arithmetic means across all three accepted workflows; categories partition every primary SQL call.

Configuration	Path SQL / ms	Payload SQL / ms	Control SQL / ms	Other SQL / ms
Off	21 / 14.296	52 / 14.367	63 / 8.853	17 / 2.781
Repaired pre-fusion / process	21 / 14.313	52 / 20.928	69 / 10.825	17 / 3.186
TTG+ / process	17 / 21.982	44 / 19.760	69 / 11.821	17 / 3.253

Payload fingerprint b0075df9: 108 returned rows and 70,163 protocol row-payload bytes

in every workflow in all three modes. TTG+ loads them in 44 calls, versus 52 off/pre-fusion.

Path discovery: native path SELECTs plus certified fused SQL. Controls include BEGIN, SET, COMMIT and DISCARD.

These contributions locate the measured cost; matched EXPLAIN diagnostics examine server planning/execution separately.

LittleX: the two fused SQL targets

Full SQL, source-bound fingerprints and per-request call references are retained in the attribution artifact.

Batched current-ability path region

3166e1168d5af494 · 4 calls/workflow · 9.680 ms mean DB sum

Requests: load_feed

Recursive certified path region

d106730720c848bc · 6 calls/workflow · 7.933 ms mean DB sum

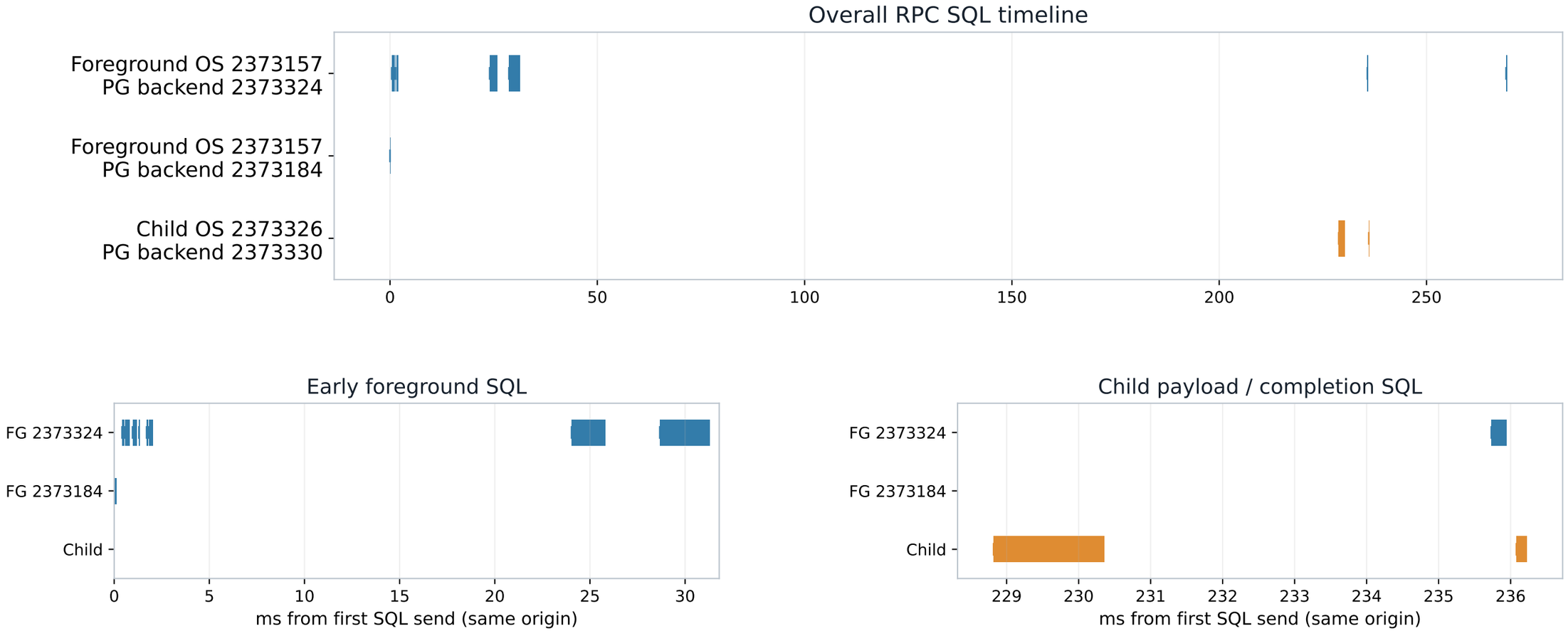
Requests: create_tweet, get_profile, load_feed

The batched query covers two current collect paths; the recursive query also appears in get_profile and create_tweet.

Protocol timing includes transport and client decode progress. The separate diagnostic recaptures exact Bind values/state.

Actual process reader: separate PID, sequential SQL intervals

LittleX TTG+ · measured repeat0 · first load_feed; actual send-to-ReadyForQuery intervals.



Bars preserve measured durations; ticks mark actual sends. Insets retain the overall timeline’s millisecond coordinates.

Real child SQL is observed; foreground/background DB intervals do not overlap in the measured workflows.

Matched LittleX diagnostic: planning and execution

One accepted recapture per mode; first and immediate repeated EXPLAIN are separate observations.

Semantic region	Observation	SQL	Planning ms	Execution ms	Rows	Shared hits
Root → Profile	First	Native	0.315	0.092	1	12
Root → Profile	First	Fused	0.742	0.184	1	22
Collect: Post + Follow	First	Native	0.943	0.278	12	55
Collect: Post + Follow	First	Fused	1.203	0.535	12	320
Root → Profile	Repeat	Native	0.244	0.070	1	9
Root → Profile	Repeat	Fused	0.617	0.094	1	22
Collect: Post + Follow	Repeat	Native	0.964	0.180	12	55
Collect: Post + Follow	Repeat	Fused	1.181	0.423	12	320

Exact recapture Bind/state in the same SERIALIZABLE transaction; census and EXPLAIN warm the buffers.

Collect native values add two independent statement roots. These server-reported times are separate from primary n3 protocol wall.

Measured plan difference: indexable target lookup

First EXPLAIN · matched collect region; plan rows are per-loop averages, distinct from returned SQL row totals.

Plan node	Index	Loops	Rows / loop	Shared hits
Native Post	anchors_pkey	4	1	8
Native Follow	anchors_pkey	8	1	16
Fused batch / Post	idx_anchors_kind_type	4	2.5	76
Fused batch / Follow	idx_anchors_kind_type	8	14.5	158

Native target nodes use the primary-key condition id = CASE … . The selected fused nodes use the kind/type index with Index Cond: kind = 'NodeAnchor'.

Collect statement-root shared hits: native pair 55; fused batch 320. Shared reads and temporary I/O are zero.

Node and parent buffer counters are inclusive. The table retains each reported counter without adding the tree.

Next work: preserve the proof, reduce small-region cost

Research hypotheses motivated by measured counts and plans; improvements have not been measured.

Indexable target joins

Reshape the fused SQL AST so target ID equality can use the primary key; retain ACL, type/module, arity and native ordering checks.

Eligibility and cost admission

Prefer fusion when the proven region coalesces enough path/payload calls. Test per-path execution for single-origin regions and unresolved Find guards.

Planning reuse

Test prepared-plan reuse for repeated fused shapes; compare planning, execution, total SQL and complete native correctness under the same regime.

Experimental regime and correctness

Same final repaired source; only the prefetch/fusion configuration differs.

Seed 42; one process reader; prefetch_limit=2048; common planner budget=32768.

Nine excluded full-workflow warmups; three measured fresh workflows per cell; balanced policy order.

Each workflow uses equivalent fresh data and fresh server state; one profile reset after setup.

Always admission model; natural within-workflow caches; no cross-trial learning or cache carry.

Native reports, ordering, Sum/Find, LittleX mutations, identities and final graph/restart checks pass.

Reviewed direct driver boundary; native process cleanup and post-response late-call attribution.

Actual client/PG backend PID intervals support process and overlap evidence; SQL logging is off.